ethical hacking unit 5 12marks | MATLAB | Ask Gemini

matlab.mathworks.com

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/ › MATLAB Drive

Editor

exp1.m × exp3.m ×

/MATLAB Drive/exp3.m

```
clc;
clear;
close all;

%% 1. Filter Specifications
Fs = 1000;              % Sampling frequency (Hz)
T = 1/Fs;               % Sampling period
N = 4;                  % Filter order
Fc = 100;               % Cutoff frequency (Hz)

Wp = 2*pi*Fc;           % Analog cutoff frequency (rad/s)

%% 2. Design Analog Butterworth Filter
[bA, aA] = butter(N, Wp, 's');

disp('Analog Butterworth Filter');
disp('Numerator coefficients:');
disp(bA);
disp('Denominator coefficients:');
disp(aA);

%% 3. Impulse Invariance Method
```

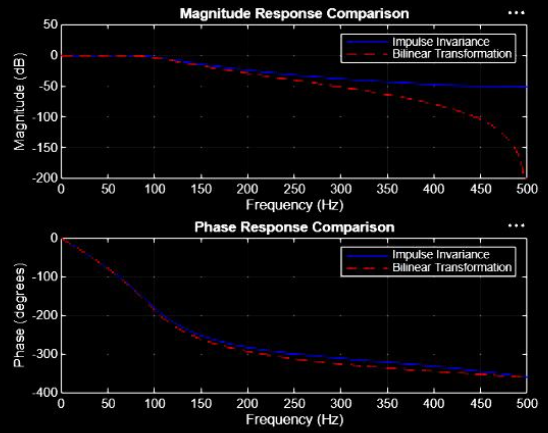
Command Window

```
Bilinear Transformation Method:
Magnitude at cutoff: -3.655 dB
Stopband attenuation: 16.930 dB

IIR filter design and comparison completed.
>>
```

Figure 2: Frequency Response Comparison

FIGURE FORMAT TOOLS LAYOUT



Read "peppers.png" and display as grayscale

exp3.m

Ask Copilot

Validate generated output before use.

Editor: 100% UTF-8 LF Script Ln 1 Col 1